Let R be a ring with 1.

1. (i) Let $h : U \to V$ be a monomorphism of R-modules. Then the following two conditions are equivalent.

   (a) A homomorphism $k : V \to Y$ must be a monomorphism whenever the composite map $U \xrightarrow{h} V \xrightarrow{k} Y$ is a monomorphism.

   (b) If $0 \neq W \subset V$, then $h(U) \cap W \neq 0$.

   The monomorphism h that satisfy the above conditions is said to be *essential.* If there is an injective R-module Q with an essential monomorphism $h : V \to Q$, then the diagram $V \xrightarrow{h} Q$ is called an *injective hull* of V.

   (ii) Prove that: any R-module V has an injective hull.

2. (i) The following holds for an idempotent e of R:

$$J(eRe) = eJ(R)e = eRe \cap J(R).$$

   (ii) Prove: $J(M_n(R)) = M_n(J(R))$. Here $M_n(R)$ means the matrix ring.

3. Prove: A submodule of a finitely generated free module over a PID is free.